Year 11 Foundation

REVISION PACK

Number 8

Name: _____

Class: _____

Number 8 · Standard form (Lesson 1.1)

Worked example: writing numbers in standard form

This question is about standard form.

(a) Write 3.8×10^4 as an ordinary number. (b) Write 0.00065 in standard form.

(a)	Step 1	10^4 = 10 000, so multiply	$3.8 \times 10\,000 = 38\,000$
(b)	Step 1	Move point to after the 6	6.5
	Step 2	Point moved 4 right: power −4	6.5×10^{-4}

Practice: writing numbers in standard form

1.1 This question is about standard form.

(a) Write 2.7×10^4 as an ordinary number.

(b) Write 0.0063 in standard form.

1.2 This question is about standard form.

(a) Write 5.04×10^5 as an ordinary number.

(b) Write 0.039 in standard form.

1.3 This question is about standard form.

(a) Write 9.1×10^3 as an ordinary number.

(b) Write 0.00805 in standard form.

1.4 This question is about standard form.

(a) Write 6.25×10^3 as an ordinary number.

(b) Write 0.00051 in standard form.

1.5 This question is about standard form.

(a) Write 1.8×10^5 as an ordinary number.

(b) Write 0.0027 in standard form.

1.6 This question is about standard form.

(a) Write 7.3×10^6 as an ordinary number.

(b) Write 0.068 in standard form.

Number 8 · Standard form (Lesson 1.1)

Worked example: multiplying in standard form

Work out $(2 \times 10^5) \times (4 \times 10^3)$. Give your answer in standard form.

Step 1	Multiply the numbers	$2 \times 4 = 8$
Step 2	Add the powers	$10^{5+3} = 10^8$
Step 3	Answer	8×10^8

Practice: multiplying in standard form

1.7 Work out $(2 \times 10^3) \times (3 \times 10^5)$. Give your answer in standard form.

1.8 Work out $(3 \times 10^6) \times (3 \times 10^2)$. Give your answer in standard form.

1.9 Work out $(1.5 \times 10^4) \times (2 \times 10^3)$. Give your answer in standard form.

1.10 Work out $(4 \times 10^2) \times (2 \times 10^7)$. Give your answer in standard form.

1.11 Work out $(2.5 \times 10^3) \times (3 \times 10^4)$. Give your answer in standard form.

1.12 Work out $(1.2 \times 10^5) \times (4 \times 10^6)$. Give your answer in standard form.

Number 8 · Standard form (Lesson 1.1)

Worked example: big and small numbers

This question is about standard form.

(a) Write 7 300 000 in standard form. (b) Write 4.5×10^{-3} as an ordinary number.

(a)	Step 1	Move point to after the 7	7.3
	Step 2	Point moved 6 left: power 6	7.3×10^{6}
(b)	Step 1	Power −3: divide by 1000	$4.5 \div 1000 = 0.0045$

Practice: big and small numbers

1.13 This question is about standard form.

(a) Write 3 800 000 in standard form.

(b) Write 2.6×10^{-2} as an ordinary number.

1.14 This question is about standard form.

(a) Write 65 000 in standard form.

(b) Write 7.1×10^{-3} as an ordinary number.

1.15 This question is about standard form.

(a) Write 920 000 000 in standard form.

(b) Write 5.08×10^{-4} as an ordinary number.

1.16 This question is about standard form.

(a) Write 410 000 in standard form.

(b) Write 1.9×10^{-2} as an ordinary number.

1.17 This question is about standard form.

(a) Write 28 000 000 in standard form.

(b) Write 6.45×10^{-3} as an ordinary number.

1.18 This question is about standard form.

(a) Write 5300 in standard form.

(b) Write 8×10^{-5} as an ordinary number.

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Worked example: prime factors

Write 84 as a product of its prime factors.

Step 1	Divide by 2	$84 = 2 \times 42$
Step 2	Divide by 2	$42 = 2 \times 21$
Step 3	Divide by 3	$21 = 3 \times 7$
Step 4	Answer	$2 \times 2 \times 3 \times 7$

Practice: prime factors

2.1 Write 90 as a product of its prime factors.

2.2 Write 126 as a product of its prime factors.

2.3 Write 150 as a product of its prime factors.

2.4 Write 48 as a product of its prime factors.

2.5 Write 200 as a product of its prime factors.

2.6 Write 132 as a product of its prime factors.

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Worked example: HCF and LCM

Find

(a) the highest common factor (HCF) of 18 and 30 (b) the lowest common multiple (LCM) of 4 and 10

(a)	Step 1	Factors of 18	$1, 2, 3, 6, 9, 18$
	Step 2	Factors of 30	$1, 2, 3, 5, 6, 10, 15, 30$
	Step 3	Biggest factor in both	6
(b)	Step 1	Multiples of 4	$4, 8, 12, 16, 20$
	Step 2	Multiples of 10	$10, 20$
	Step 3	Smallest multiple in both	20

Practice: HCF and LCM

2.7 Find

(a) the highest common factor (HCF) of 20 and 30 (b) the lowest common multiple (LCM) of 4 and 6

2.8 Find

(a) the highest common factor (HCF) of 16 and 40 (b) the lowest common multiple (LCM) of 6 and 9

2.9 Find

(a) the highest common factor (HCF) of 27 and 45 (b) the lowest common multiple (LCM) of 8 and 12

2.10 Find

(a) the highest common factor (HCF) of 28 and 42 (b) the lowest common multiple (LCM) of 5 and 7

2.11 Find

(a) the highest common factor (HCF) of 32 and 48 (b) the lowest common multiple (LCM) of 9 and 12

2.12 Find

(a) the highest common factor (HCF) of 15 and 35 (b) the lowest common multiple (LCM) of 10 and 15

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Worked example: prime factors and the HCF

This question is about 90 and 48

(a) Write 90 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 90 and 48

(a)	**Step 1**	Divide by primes	$90 = 2 \times 45 = 2 \times 3 \times 15$
	Step 2	Answer	$2 \times 3 \times 3 \times 5$
(b)	**Step 1**	Prime factors of 48	$2 \times 2 \times 2 \times 2 \times 3$
	Step 2	Primes in both	$2 \times 3 = 6$

Practice: prime factors and the HCF

2.13 This question is about 72 and 54

(a) Write 72 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 72 and 54

2.14 This question is about 120 and 90

(a) Write 120 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 120 and 90

2.15 This question is about 126 and 84

(a) Write 126 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 126 and 84

2.16 This question is about 100 and 75

(a) Write 100 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 100 and 75

2.17 This question is about 60 and 45

(a) Write 60 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 60 and 45

2.18 This question is about 98 and 70

(a) Write 98 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 98 and 70

Number 8 · Ratio (Lesson 1.4)

Worked example: simplifying ratios

Write the ratio 20 : 35 in its simplest form.

Step 1	HCF of 20 and 35	5
Step 2	Divide both by 5	$20 \div 5 = 4,\ 35 \div 5 = 7$
Step 3	Answer	$4 : 7$

Practice: simplifying ratios

3.1 Write the ratio 15 : 25 in its simplest form.

3.2 Write the ratio 28 : 42 in its simplest form.

3.3 Write the ratio 36 : 48 in its simplest form.

3.4 Write the ratio 30 : 45 in its simplest form.

3.5 Write the ratio 16 : 40 in its simplest form.

3.6 Write the ratio 27 : 63 in its simplest form.

Number 8 · Ratio (Lesson 1.4)

Worked example: sharing in a ratio

Ali and Sara share 56 KD in the ratio 3 : 4. How much does each person get?

Step 1	Add the parts	$3 + 4 = 7$
Step 2	One part	$56 \div 7 = 8$
Step 3	Ali	$3 \times 8 = 24$ KD
Step 4	Sara	$4 \times 8 = 32$ KD

Practice: sharing in a ratio

3.7 Omar and Mona share 60 KD in the ratio 2 : 3. How much does each person get?

3.8 Fahad and Laila share 72 KD in the ratio 5 : 3. How much does each person get?

3.9 Noor and Ahmad share 90 KD in the ratio 7 : 2. How much does each person get?

3.10 Yousef and Huda share 40 KD in the ratio 3 : 5. How much does each person get?

3.11 Sara and Ali share 84 KD in the ratio 5 : 7. How much does each person get?

3.12 Maha and Faisal share 100 KD in the ratio 3 : 7. How much does each person get?

Number 8 · Ratio (Lesson 1.4)

Worked example: recipes

In a recipe, the ratio of flour to sugar is 4 : 3. Mona uses 280 g of flour. How much sugar should Mona use?

Step 1	One part	$280 \div 4 = 70$
Step 2	Sugar is 3 parts	$3 \times 70 = 210$
Step 3	Answer	$210\ \text{g}$

Practice: recipes

3.13 In a recipe, the ratio of flour to butter is 3 : 2. Ali uses 240 g of flour. How much butter should Ali use?

3.14 In a recipe, the ratio of flour to sugar is 7 : 2. Sara uses 420 g of flour. How much sugar should Sara use?

3.15 In a recipe, the ratio of juice to water is 2 : 5. Omar uses 150 ml of juice. How much water should Omar use?

3.16 In a recipe, the ratio of flour to butter is 4 : 1. Huda uses 360 g of flour. How much butter should Huda use?

3.17 In a recipe, the ratio of sugar to flour is 3 : 5. Fahad uses 120 g of sugar. How much flour should Fahad use?

3.18 In a recipe, the ratio of flour to sugar is 5 : 4. Laila uses 450 g of flour. How much sugar should Laila use?

Number 8 · Ratio (Lesson 1.4)

Worked example: finding the total

The ratio of red beads to blue beads in a box is 2 : 5. There are 14 red beads. How many red beads and blue beads are in the box altogether?

Step 1	One part	$14 \div 2 = 7$
Step 2	Total parts	$2 + 5 = 7$
Step 3	Total parts × one part	$7 \times 7 = 49$

Practice: finding the total

3.19 The ratio of red beads to blue beads in a box is 3 : 5. There are 18 red beads. How many red beads and blue beads are in the box altogether?

3.20 The ratio of green counters to yellow counters in a box is 4 : 7. There are 28 green counters. How many green counters and yellow counters are in the box altogether?

3.21 The ratio of white tiles to black tiles in a box is 2 : 9. There are 12 white tiles. How many white tiles and black tiles are in the box altogether?

3.22 The ratio of pens to pencils in a box is 5 : 3. There are 35 pens. How many pens and pencils are in the box altogether?

3.23 The ratio of apples to oranges in a box is 3 : 8. There are 27 apples. How many apples and oranges are in the box altogether?

3.24 The ratio of red cubes to blue cubes in a box is 4 : 5. There are 32 red cubes. How many red cubes and blue cubes are in the box altogether?